

Spring-Mass System & Torsional Pendulum



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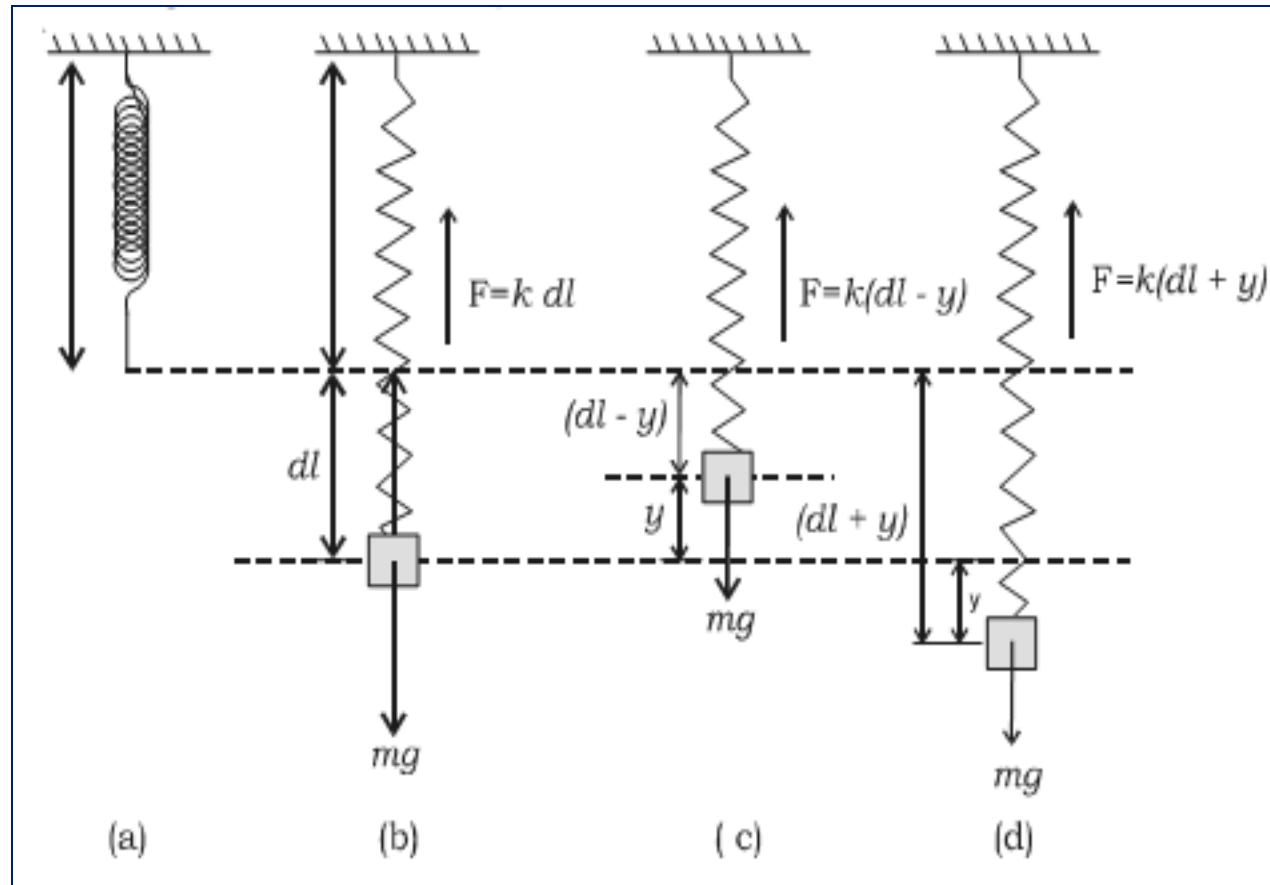
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Whatever we will discuss

- ✓ Vertical oscillations of a spring
- ✓ Combinations of spring
- ✓ Torsional Pendulum
- ✓ Related mathematical problems

Vertical oscillations of a spring

The restoring force
 $F = k \, dl$; $k \, dl = mg$

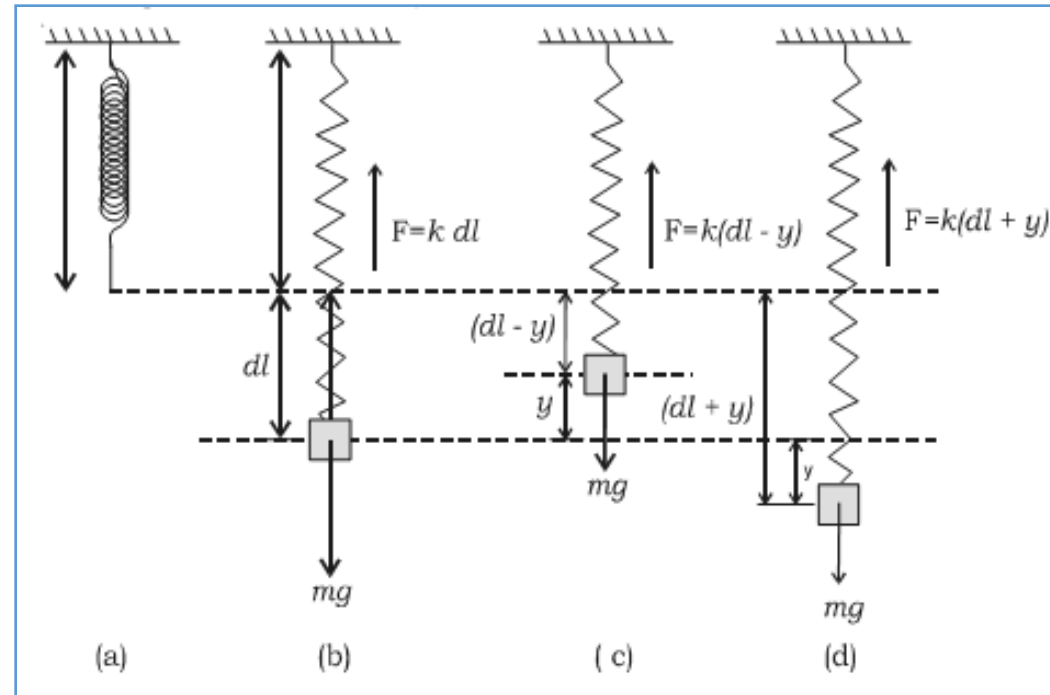


The extension of the spring is $(dl - y)$. The upward force exerted on the body is $k (dl - y)$ and the downward force is mg . Hence the resultant force F on the body is $F = k (dl - y) - mg = -ky$

Vertical oscillations of a spring

If the total extension produced is $(dl + y)$ as in Fig.-(d) the restoring force on the body acts upwards, which is

$$F = k(dl + y)$$



So, the increase in the upward force on the spring is

$$k(dl + y) - mg = ky$$

Vertical oscillations of a spring

As the force is proportional to the displacement of the body from its equilibrium position and acts in the opposite direction to that of displacement, the restoring force is $-ky$. Thus, the motion is Simple Harmonic Motion.

According to Newton's second law of motion,

$$F = m \frac{d^2 y}{dt^2} = -k y$$

$$\text{or, } m \frac{d^2 y}{dt^2} + k y = 0$$

$$\text{or, } \frac{d^2 y}{dt^2} + \frac{k}{m} y = 0$$

The equation is similar to the differential equation of simple harmonic motion

$$\frac{d^2 y}{dt^2} + \omega^2 y = 0$$

Hence the motion of a mass suspended from a spiral spring is simple harmonic.

Vertical oscillations of a spring

$$mg = k \, dl$$

$$\frac{m}{k} = \frac{dl}{g}$$

Therefore time period

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{dl}{g}} \quad \text{As, } \omega^2 = \frac{k}{m}; \quad \omega = \sqrt{\frac{k}{m}}$$

Frequency

$$f = \frac{1}{2\pi} \sqrt{\frac{g}{dl}}$$

When two springs are connected in parallel

If F_1 and F_2 are the restoring forces

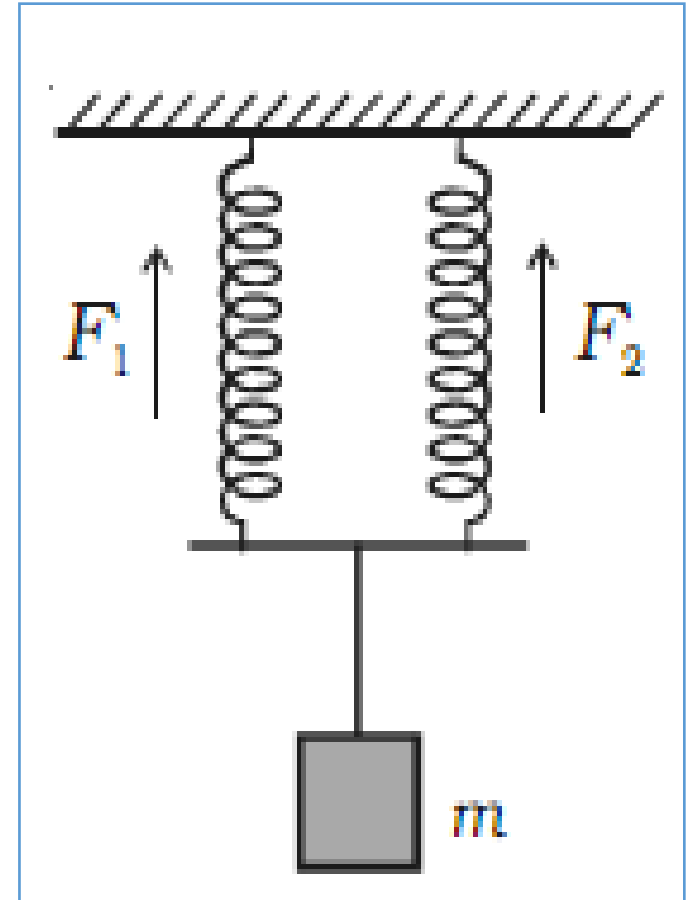
$$F_1 = -k_1 y, \quad F_2 = -k_2 y$$

$$\text{Total restoring force} = (F_1 + F_2) = - (k_1 + k_2) y$$

$$T = 2\pi \sqrt{\frac{m}{k_1 + k_2}}$$

$$\text{If } k_1 = k_2 = k \quad T = 2\pi \sqrt{\frac{m}{2k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$



When two springs are connected in series

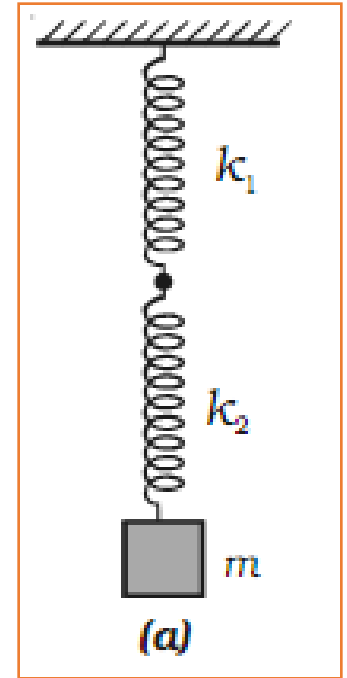
$$F = -k_1 y_1 \quad ; \quad F = -k_2 y_2$$

$$\text{Total extension } y = y_1 + y_2 = -F \left[\frac{1}{k_1} + \frac{1}{k_2} \right]$$

We know that $F = -ky$

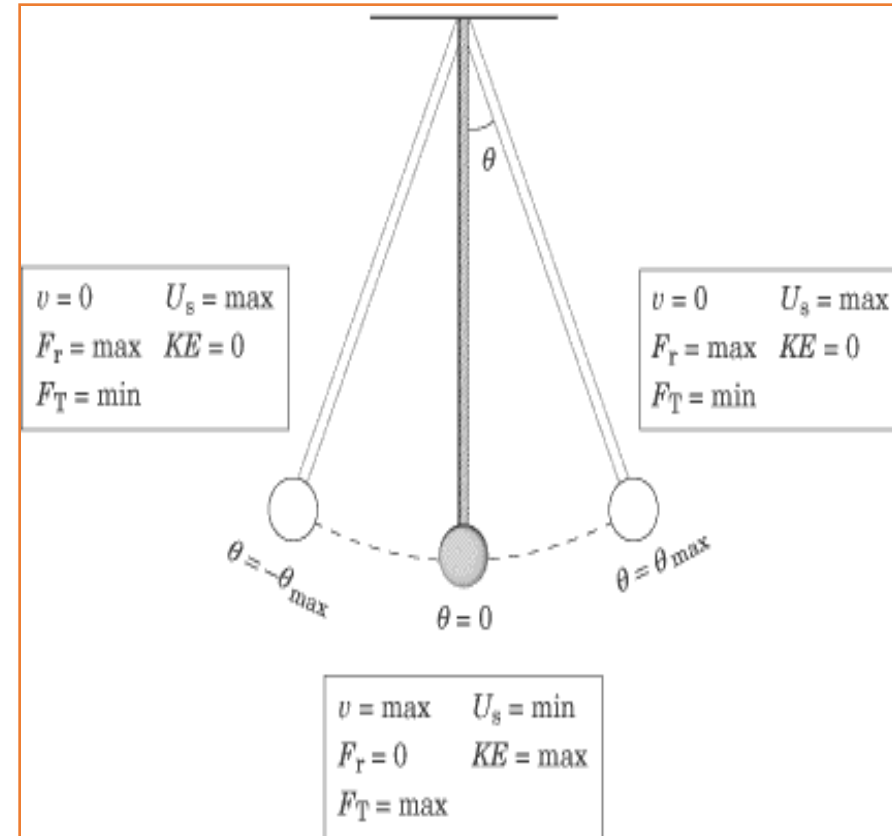
$$-\frac{F}{k} = -F \left[\frac{1}{k_1} + \frac{1}{k_2} \right]$$

$$k = \frac{k_1 k_2}{k_1 + k_2}$$



Pendulum

A pendulum is a weight suspended from a pivot so that it can swing freely.

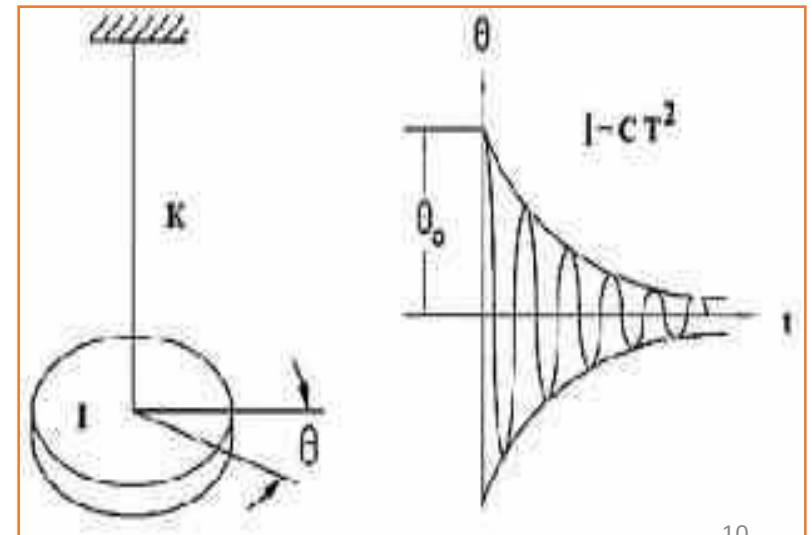
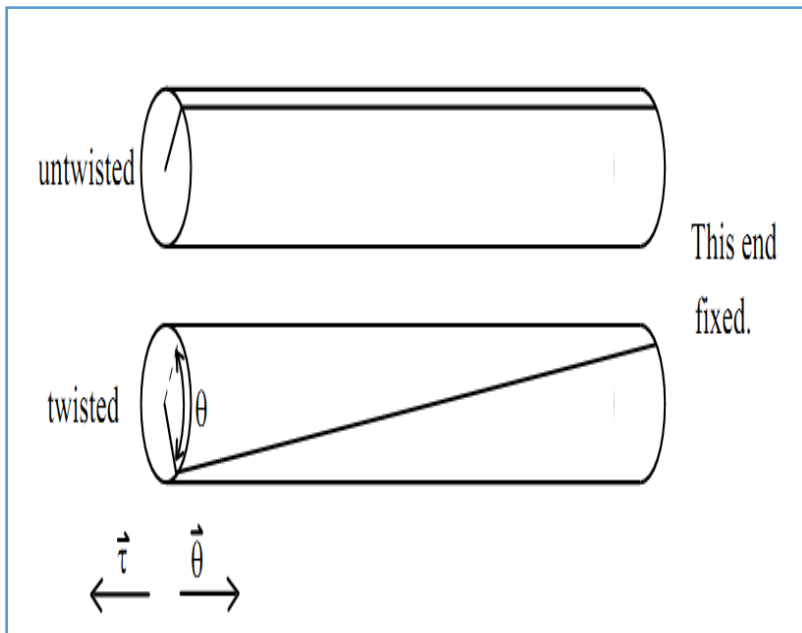


Torsional Pendulum

A torsional pendulum, or torsional oscillator, consists of a disk-like mass suspended from a thin rod or wire.

For a small twists restoring torque is found to be proportional to the angular displacement. $\tau = -\kappa\theta \dots \dots \dots (1)$

Like as, $F = -kx$ for a spring



Torsional Pendulum

The equation of motion for such a system is based on the angular form of Newton's second law

$$\tau = I\alpha = I \frac{d^2\theta}{dt^2} \dots \dots \dots (2) \quad \text{Like as, } F = ma = m \frac{d^2x}{dt^2}$$

From the above equations (1) and (2), we can write

$$-\kappa\theta = I \frac{d^2\theta}{dt^2}$$

$$\frac{d^2\theta}{dt^2} + \frac{\kappa}{I} \theta = 0 \quad \text{Like as } \frac{d^2x}{dt^2} + \frac{k}{m} x = 0$$

Torsional Pendulum

The solution of the differential equation

$$\theta = \theta_m \sin(\omega t + \varphi) \qquad x = x_m \sin(\omega t + \varphi)$$

$$\omega = \sqrt{\frac{\kappa}{I}}$$

$$\omega = \sqrt{\frac{k}{m}}$$

So the time period of oscillations

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{\kappa}}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

Problem: A particle performs simple harmonic motion given by the equation $y = 20 \sin(\omega t + \alpha)$. If the time period is 30 seconds and the particle has a displacement of 10 cm at $t = 0$ seconds, find (i) epoch, (ii) the phase angle at $t = 5$ seconds and (iii) the phase difference between two positions of the particle 15 seconds apart.

Mathematical Problems

Home Task: Example 1.1 to Example 1.16

Physics for Engineers (Part-1) -Dr. Gias Uddin Ahmed; pp: 10-45



Thank You
Thanks for your attention